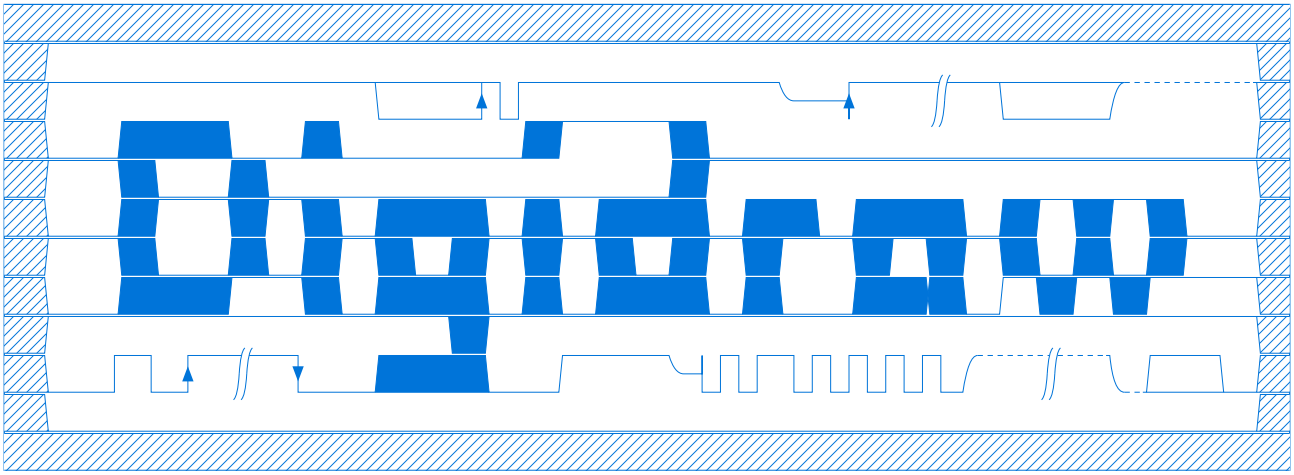


Created by Joel von Rotz

Source Code on Codeberg

DIGIDRAW is a Typst package to help with drawing digital timing diagrams. The syntax tries to be like from *WaveDrom*, although is slightly different in some areas. In addition, the package supports more customization such as configuring colors, fonts and others.

The package uses CETZ to draw everthing.

> **Caution**
> Not all symbol combinations have been implemented so far (probably) or they differ from WaveDrom, so expect some undefined or unexpected behaviour.
>
> If they are straying too far away from the timing diagram standard, please report them in the package repository: *https://codeberg.org/joelvonrotz/typst-digidraw/*

Contents

I \ Installation

I.1 From Universe (aka. Stable Version)

To import the Universe version, insert following snippet

```
#import "@preview/digidraw:0.9.3" as dd
```

I.2 From Repository (aka. Development Version)

The repository will always be the *development* version of DIGIDRAW. This will also include the latest changes, but might break here and there.

I. Clone the Repository into your project (for example into a `./packages/` folder or at root level).

```
cd /path/to/project/
git clone https://codeberg.org/joelvonrotz/typst-digidraw.git
```

OR if your project is a git project, you can add it as a submodule, saving a little bit of space in your repository:

```
cd /path/to/project/
git submodule add https://codeberg.org/joelvonrotz/typst-digidraw.git
```

II. Include the package into your document

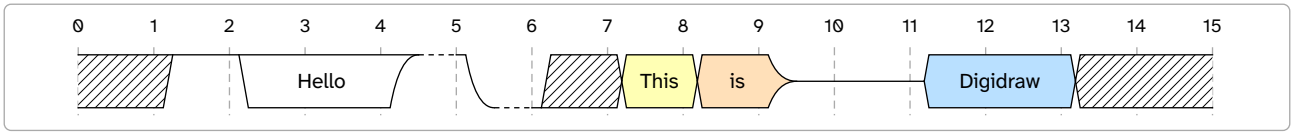
```
#import "./path/to/typst-digidraw/lib.typ" as dd
```

II \ Usage

II.1 Drawing Diagrams

Drawing a diagram is easy:

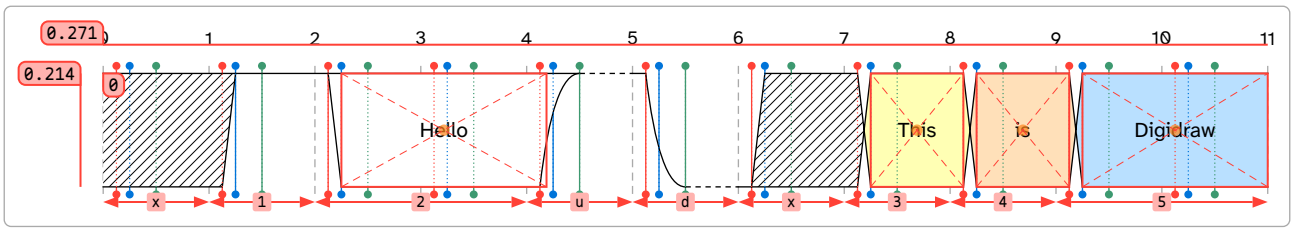
```
#dd.wave(
  (signal: (
    (wave: "x12.udx34z.5.x.",data: "Hello This is Digidraw"),)
  )
)
```



II.2 Debugging Diagrams

You can turn on Debug mode by setting `debug: true` in the `wave()` function. This shows where and which symbol is placed and how long said symbol is. Additionally the `step1`, `step2` and `step3` parameters are shown with respective coloured vertical lines. Debug mode also shows the exact placement of bus labels using a bounding box and a center point.

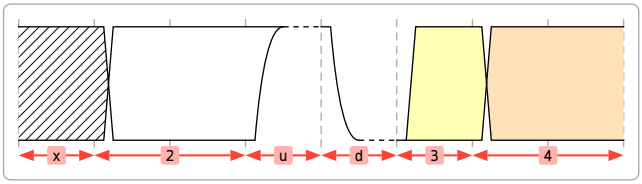
```
#dd.wave(
  (signal: (
    (wave: "x12.udx345.", data: "Hello This is Digidraw"),
  )),
  debug: true
)
```



This looks quite cluttered and is not the actual recommended way to use this feature. Instead you can pass one or more of `"symbols"`, `"labels"`, `"coordinates"` and `"steps"` to show the respective information.

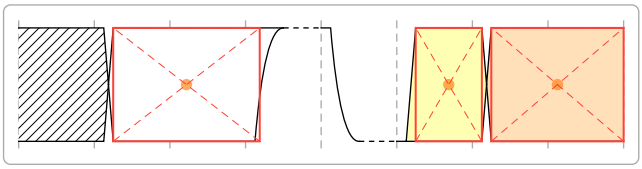
- `"symbols"` – shows the arrows below the respective wave to show which shape belongs to which symbol.

```
#dd.wave(
  (signal: ((wave: "x2.ud34."),)),
  tick-format: none,
  debug: "symbols"
)
```



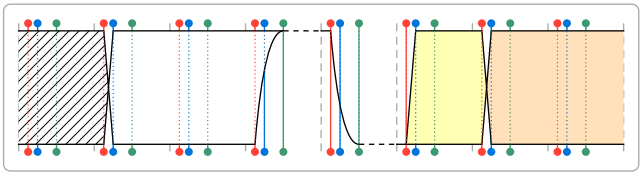
- `"labels"` – shows the area in which labels can be placed. By default labels are placed `horizon+center` in the respective box. This can be changed per entry by enclosing the label in a `#align(..)`.

```
#dd.wave(
  (signal: ((wave: "x2.ud34."),)),
  tick-format: none,
  debug: "labels"
)
```



- `"steps"` – shows the «steps» of each symbol. These are the x coordinates (per symbol) used for transition (i.e. `0`→`1`) and bezier curves (i.e. `2`→`u`).

```
#dd.wave(
  tick-format: none,
  (signal: ((wave: "x2.ud34."),)),
  debug: "steps"
)
```

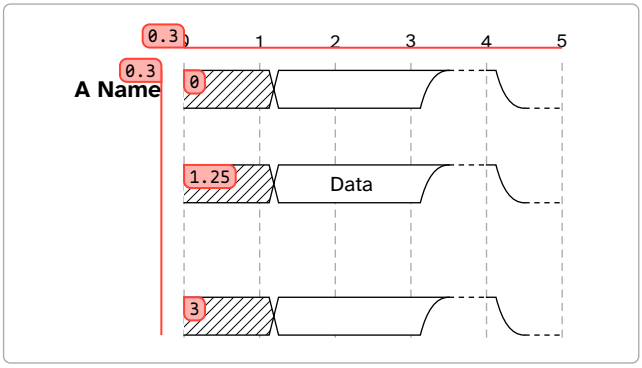


- `"coordinates"` – shows the vertical coordinates of each wave, which helps when using the `others` parameter.

Important

Coordinates are given in **`cetz`** units and are relative to `symbol-width`. So $l_{\text{actual}} = l_{\text{relative}} \cdot$ `symbol-width` can be used to get the absolute lengths from relative coordinates.

```
#dd.wave(
  symbol-width:  1cm,
  symbol-height: 5mm,
  wave-gutter: 7.5mm,
  (signal: (
    (wave: "x2.ud", name: "A Name"),
    (wave: "x2.ud", data: ([Data],)),
    (:),
    (wave: "x2.ud"))
  ),
  debug: "coordinates"
)
```



II.3 Data Structure

Let's talk about wave data structures! The data syntax is the one from WaveDrom. You can copy *WaveDrom* scripts and insert those into your document. Well, almost... there are certain features such as groups that are not yet implemented and are thus ignored.

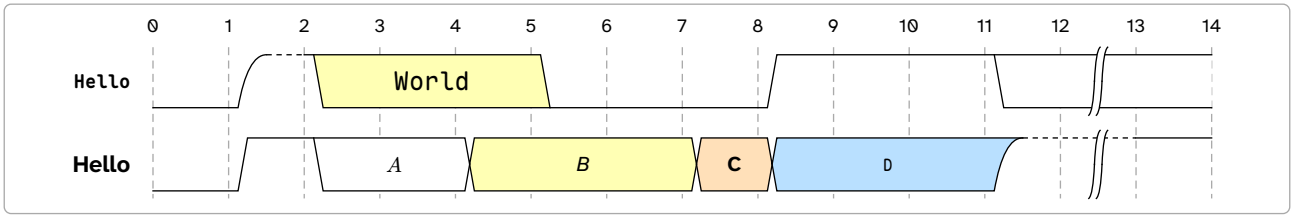
There's two ways on how to define the wave data: as dictionaries or as JSON structures

II.3.a Data-Format: Dictionaries

This allows for fancier styling labels. `.wave` and `.data` can contain *`content`* . Raw'em, Strong'em, Emph'em, do what you want, as long as it turns into a content.

```
#let data = (signal: (
  (wave: "0u3..0..1..2|.", name: `Hello`, data: (text(1.5em,`World`),)),
  (wave: "012.3..45..u|1", name: "Hello", data: ($A$, [_B_], [*C*], `D`))
))

#dd.wave(data)
```

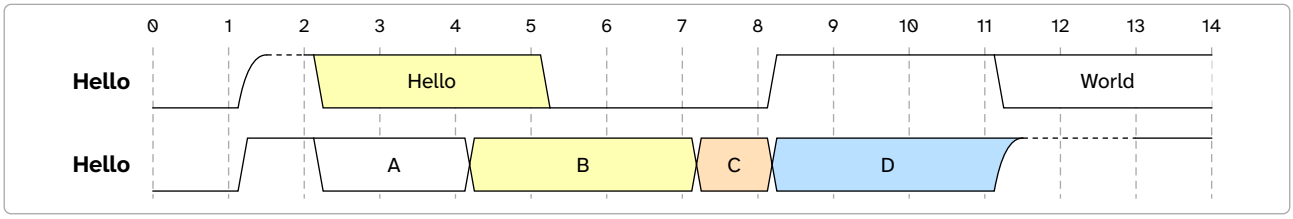


II.3.b Data-Format: JSON

Easier when copying from WaveDrom with the downside of not having much styling options compared to the dictionary based structure.

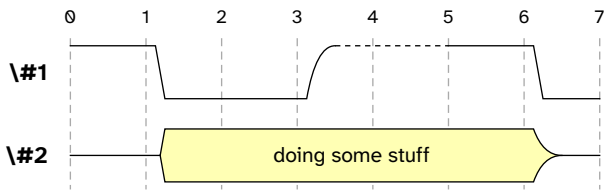
```
#let data = json(bytes(`{"signal": [
  {"wave": "0u3..0..1..2..", "name": "Hello", "data": ["Hello","World"]},
  {"wave": "012.3..45..u.1", "name": "Hello", "data": "A B C D"}
]}`.text)) // or json(read("file.json"))

#dd.wave(data)
```



II.3.c Structure

```
#let data = (
  signal: (
    ( wave: "10.u.10", name: "\#1" ),
    (
      wave: "z3....z",
      name: "\#2",
      data: ([doing some stuff],)
    ),
  )
)
#dd.wave(data)
```



.signal `array` of `dictionary`

A list/array containing the wave objects.

.wave `string`

The wave's contents built using the *Symbols*.

Example: `"10..u.10"`

.name `string` or `content`

The name of the wave.

Example: `"Mike"` or `[Mike]`

.data `string` (JSON, Typst) or `array` of `content` (Typst only) or `array` of `string` (JSON, Typst)

When buses are defined in the wave, labels can be inserted. The index of the label defines its position: first entry goes into the first bus, second in the second one, etc. If data is a string, labels are split by spaces.

i.e. `"Hello World"` is equal to `("Hello","World")`

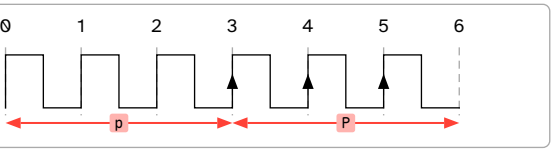
III \ Symbols

III.1 Clocks

III.1.a Normal Clocks `p` `P`

Capitalizing the letters adds an arrow mark to the line pointing up.

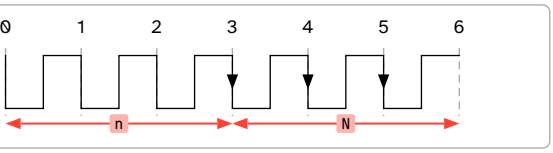
```
#dd.wave(
  (signal: ((wave: "pppPPP",),)),
  debug: "symbols"
)
```



III.1.b Inverted Clocks `n` `N`

Capitalizing the letters adds an arrow mark to the line pointing up.

```
#dd.wave(
  (signal: ((wave: "nnnNNN",),)),
  debug: "symbols"
)
```

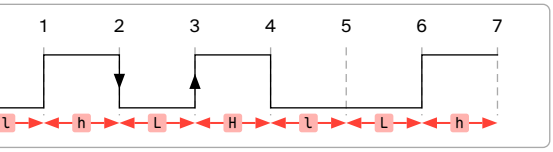


III.2 Wires

III.2.a Straight Signals `l` `L` `h` `H`

Capitalizing the letters adds an arrow mark to the line pointing up.

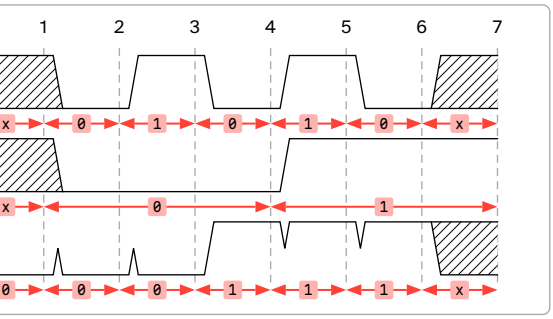
```
#dd.wave(
  (signal: ((wave: "lhLHlLh",),)),
  debug: "symbols"
)
```



III.2.b Flanked Signals `0` `1`

Using `0` and `1` instead of `l`/`L` and `h`/`H` inserts flanked/sloped signals.

```
#dd.wave(
  (signal: (
    (wave: "x01010x",),
    (wave: "x0..1..",),
    (wave: "000111x",),
  )),
  debug: "symbols"
)
```



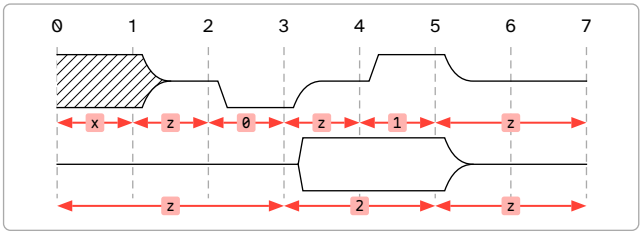
III.2.c High-Impedance Signal `z`

The high impedance signal introduces curved transitions from any signal to high impedance.

Note multiples of the symbols in a sequence are treated as one long one

Examples `"zzz2.zz"` → `"z..2.z."`

```
#dd.wave(
  (signal: (
    (wave: "xz0z1z.",),
    (wave: "zzz2.zz",),
  )),
  debug: "symbols"
)
```



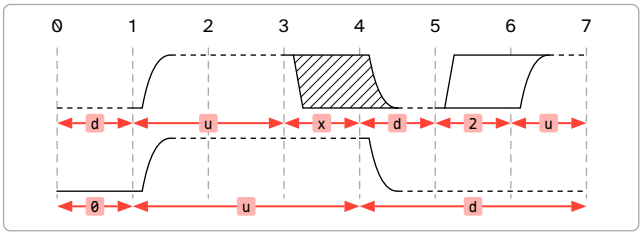
III.2.d Transition Edges `u` `d`

If the transition has of unknown duration/delay, a transition edge can be used.

Note multiples of the symbols in a sequence are treated as one long one

Examples `"uuuddduuu"` → `"u..d..u.."`

```
#dd.wave(
  (signal: (
    (wave: "du.xd2u",),
    (wave: "0uuuddd",),
  )),
  debug: "symbols"
)
```



III.3 Buses

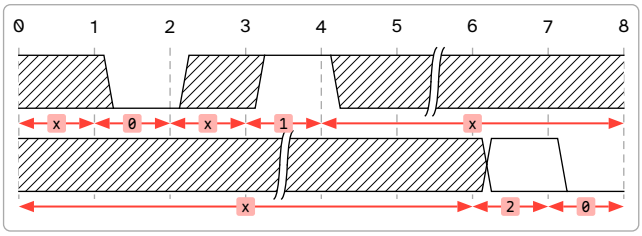
III.3.a Don't Care `x`

Indicates a section, where the signal value is not important, aka. *don't care*.

Note multiples of the symbols in a sequence are treated as one long one

Examples `"xxx|xxx20"` → `"x..|...20"`

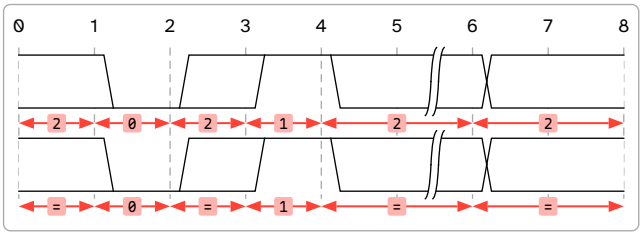
```
#dd.wave(
  (signal: (
    (wave: "x0x1x|x.",),
    (wave: "xxx|xx20",),
  )),
  debug: "symbols"
)
```



III.3.b Simple Bus `=` `2`

= and 2 are similar in that they have the same bus color. That's it, nothing special, really!

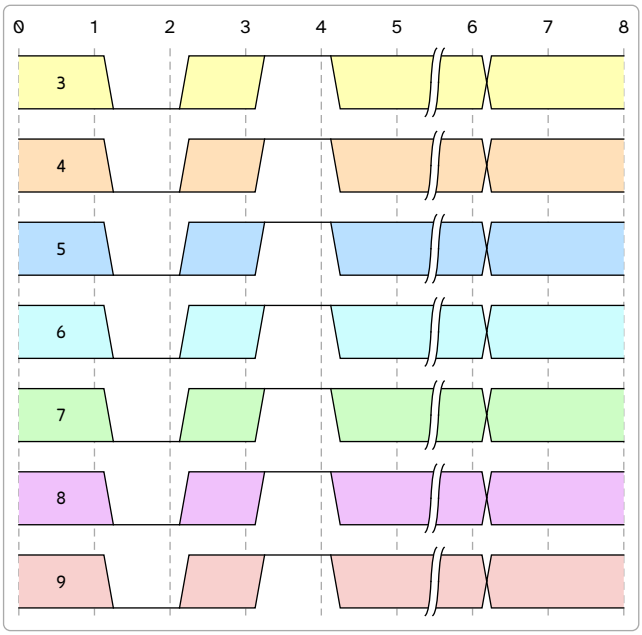
```
#dd.wave(
  (signal: (
    (wave: "20212|2.",),
    (wave: "=0=1=|=.",),
  )),
  debug: "symbols"
)
```



III.3.c Coloured Buses 3 4 5 6 7 8 9

The databus can be also colored using numbers "3" through "9".

```
#dd.wave(
  (signal: (
    (wave: "30313|3.", data: (`3`,)),
    (wave: "40414|4.", data: (`4`,)),
    (wave: "50515|5.", data: (`5`,)),
    (wave: "60616|6.", data: (`6`,)),
    (wave: "70717|7.", data: (`7`,)),
    (wave: "80818|8.", data: (`8`,)),
    (wave: "90919|9.", data: (`9`,)),
  )),
)
```

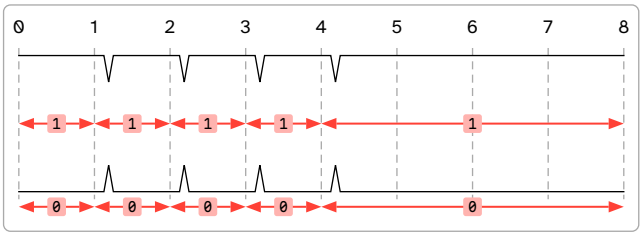


III.4 Extender .

The extender can be used to extend signals. If a signal doesn't look right, you might have to replace some of the symbols with the extender. And if that doesn't work, report it!

For example, while a "1111" inserts high signals with impulses, "1..." stretches.

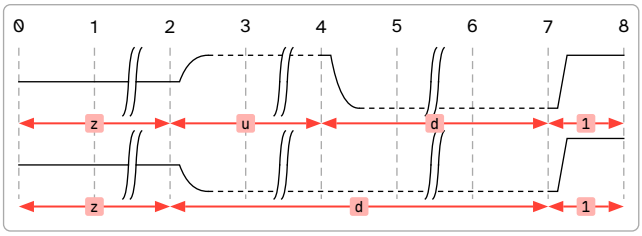
```
#dd.wave(
  (signal: (
    (wave: "11111..."),
    (wave: "00000..."),
  )),
  debug: "symbols"
)
```



III.5 Time Skip |

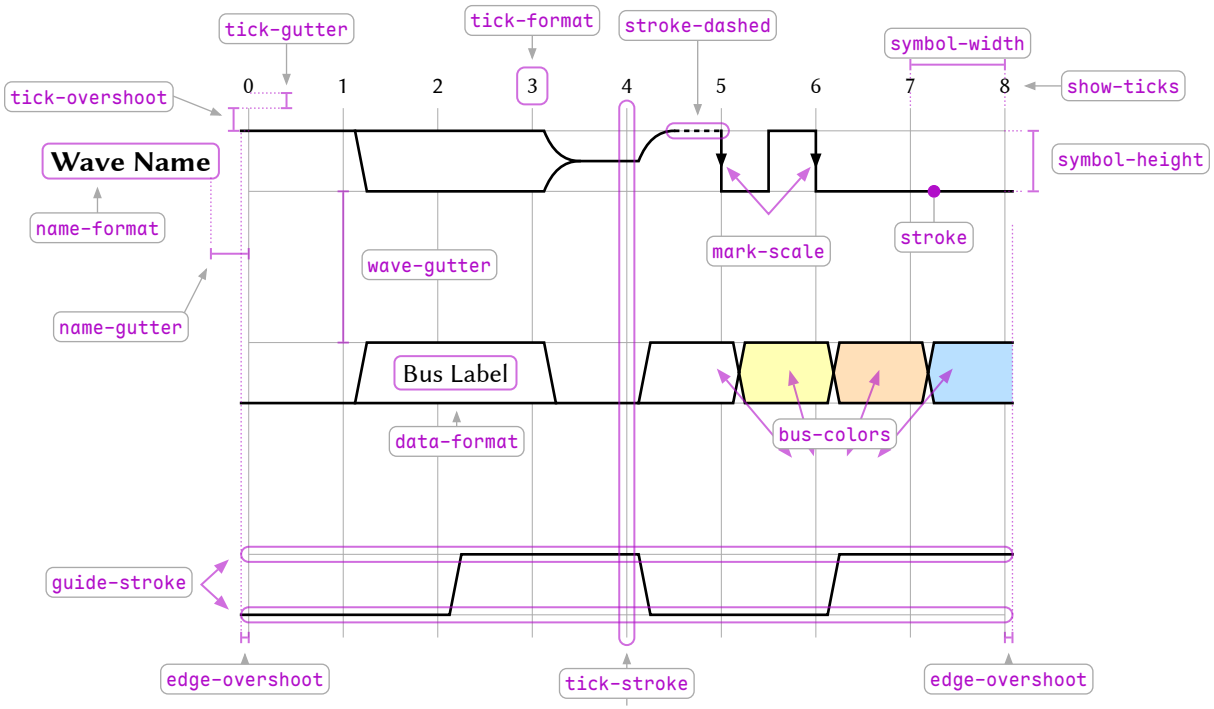
To insert a timeskip to indicate the passing of a long time, insert the symbol "|".

```
#dd.wave(
  (signal: (
    (wave: "z|u|d|.1"),
    (wave: "z|d|d|.1"),
  )),
  debug: "symbols"
)
```



IV \ API Reference

A lot of things in a timing diagram can be customized, ranging from sizing and gutter spacing to coloring and text formatting. Below the most important and useful parameters are shown and their graphical effect.



Note values assigned to `step1`, `step2`, `step3` and `bezier-controlpoint` are **X**-axis components. The Y-axis components are assigned by the `#wave` function. The example above has the values $10\%, 30\%, 50\%, 20\%$ respectively.

IV.1 wave

The one and only

IV.1.a Parameters

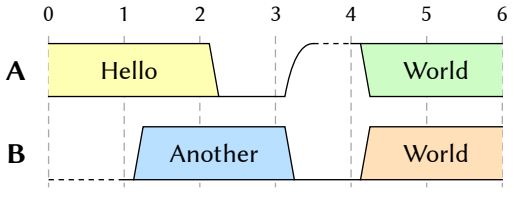
```
wave(
  data: dictionary,
  debug: string array bool,
  symbol-width: length,
  symbol-height: length ratio float,
  step1: ratio float length,
  step2: ratio float length,
  step3: ratio float length,
  bezier-controlpoint: ratio float length,
  edge-overshoot: ratio float length,
  wave-gutter: ratio float length,
  name-gutter: ratio float length,
  s-spacing: ratio float length,
  s-width: ratio float length,
  s-outside: ratio float length,
  mark-scale: float,
  stroke: stroke,
  stroke-dashed: stroke,
  show-ticks: bool function,
  tick-gutter: ratio float length,
  tick-overshoot: ratio float length,
  tick-format: none function,
  tick-stroke: none stroke function,
  guide-stroke: none stroke dictionary,
  name-format: function,
  data-format: function,
  bus-colors: dictionary,
  others: cetz
)
```

data `dictionary`

The wave data à la WaveDrom syntax.

```
#let data = (signal: (
  (wave: "3.0u7.", name: "A", data: "Hello
World"),
  (wave: "d5.04.", name: "B", data:
([Another],[World],)),
))

#dd.wave(data)
```



debug `string` or `array` or `bool`

Passing one (as a string) or more (as an array) of `"symbols"`, `"steps"`, `"coordinates"` and `"labels"`, displays the respective debugging information. Passing `true` enables all informations at once (**not recommended**). `false` disables it.

`"symbols"` Displays where each symbol is located and their length

`"steps"` Displays step1, step2 and step3 via vertical lines (as these parameters are applied to every symbol)

`"coordinates"` Shows coordinates (in `cetz` units), which comes in handy when placing elements via others into the diagram.

`"labels"` Renders a bounding box and a center point of where a bus label/data can be placed. Using `align(..)` defines the location (see default of data-format on how it is used)

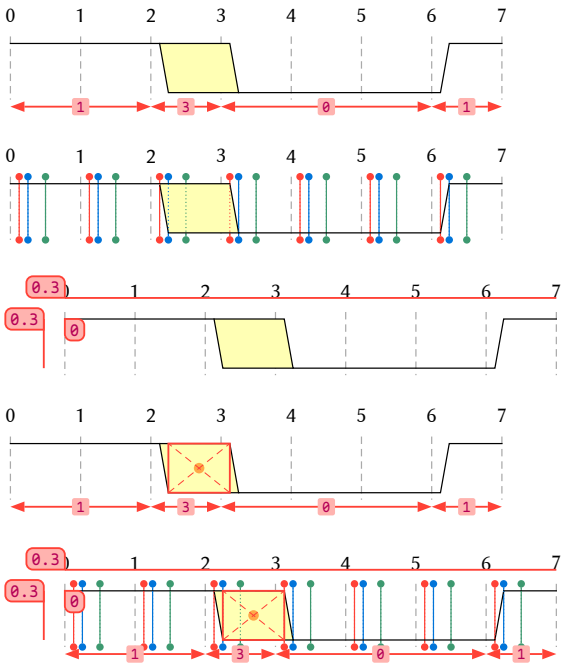
```
#dd.wave(debug: "symbols",
  (signal: ((wave: "1.30..1"),)))

#dd.wave(debug: "steps",
  (signal: ((wave: "1.30..1"),)))

#dd.wave(debug: "coordinates",
  (signal: ((wave: "1.30..1"),)))

#dd.wave(debug: ("labels", "symbols"),
  (signal: ((wave: "1.30..1"),)))

// not recommended:
#dd.wave(debug: true,
  (signal: ((wave: "1.30..1"),)))
```



Default: `false`

symbol-width `length`

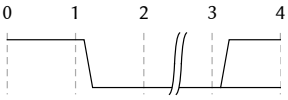
The width of one symbol and also the reference size of the diagram. All other length-based parameters are scaled to `cetz` units using this parameter ($l_{\text{relative}} = l_{\text{absolute}} / \texttt{symbol-width}$)

```
#let data = (signal: ((wave: "10|2"),))

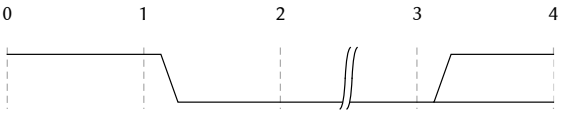
*symbol-width: 1cm* (default)
#dd.wave(data)

*symbol-width: 2cm*
#dd.wave(data, symbol-width: 2cm)
```

symbol-width: 1cm (default)



symbol-width: 2cm



Default: `1cm`

symbol-height `length` or `ratio` or `float`

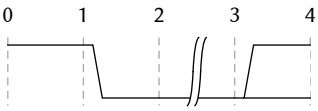
The height of one symbol or one wave, but applied to all waves. Float and ratio values are treated as relative to symbol-width.

```
#let data = (signal: ((wave: "10|2"),))

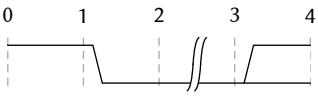
*symbol-height: 7mm* (default)
#dd.wave(data)

*symbol-height: 50%*
#dd.wave(data, symbol-height: 50%)
```

symbol-height: 7mm (default)



symbol-height: 50%



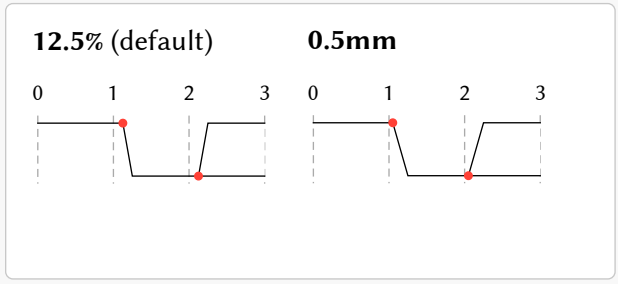
Default: `7mm`

step1 `ratio` or `float` or `length`

First inset at which a transition change starts (i.e. `"0"` → `"1"`). The value is the `X`-axis component, while the `Y`-axis component is set by `wave`. The example below highlights the inset with red circles (not actually drawn in real-use).

```
#let data = (signal: ((wave: "102"),))

#stack(dir: ltr, spacing: 5mm, [
  *12.5%* (default)
  #dd.wave(data)
],[
  *0.5mm*
  #dd.wave(data, step1: 0.5mm)
])
```



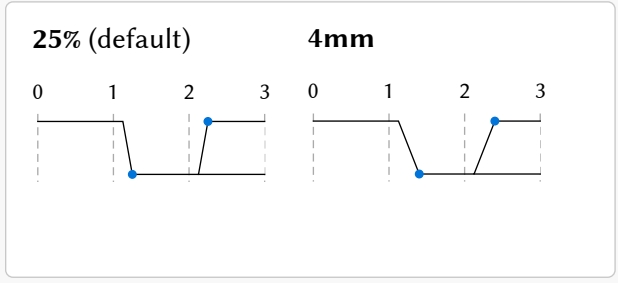
Default: `12.5%`

step2 `ratio` or `float` or `length`

Second inset at which a transition change ends (i.e. `"0"` → `"1"`). The value is the `X`-axis component, while the `Y`-axis component is set by `wave`. The example below highlights the inset with blue circles (not actually drawn in real-use).

```
#let data = (signal: ((wave: "102"),))

#stack(dir: ltr, spacing: 5mm, [
  *25%* (default)
  #dd.wave(data)
],[
  *4mm*
  #dd.wave(data, step2: 4mm)
])
```



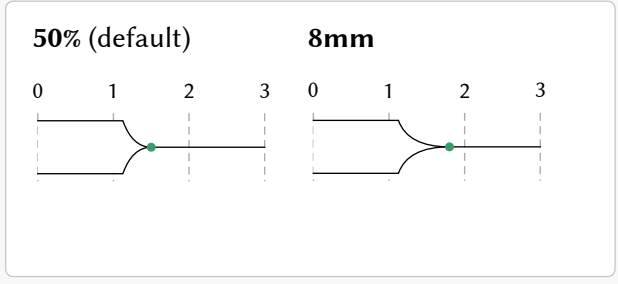
Default: `25%`

step3 `ratio` or `float` or `length`

Third inset at which a tristate-like transition ends (i.e. `"1"` → `"z"`). The value is the `X`-axis component, while the `Y`-axis component is set by `wave`. The example below highlights the inset with green circles (not actually drawn in real-use).

```
#let data = (signal: ((wave: "=z."),))

#stack(dir: ltr, spacing: 5mm, [
  *50%* (default)
  #dd.wave(data)
],[
  *8mm*
  #dd.wave(data, step3: 8mm)
])
```



Default: `50%`

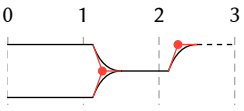
bezier-controlpoint `ratio` or `float` or `length`

Third inset at which a tristate-like transition ends (i.e. `"1"` → `"z"`). The value is the `X`-axis component, while the `Y`-axis component is set by `wave`. The example below highlights the bezier control point and its start and end points (controlled via `step1` and `step3` respectively) with red circles and red lines (not actually drawn in real-use).

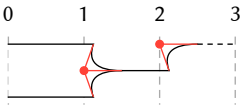
```
#let data = (signal: ((wave: "=zu"),))

#stack(dir: ltr, spacing: 5mm, [
  *25%* (default)
  #dd.wave(data)
],[
  *0%*
  #dd.wave(data, bezier-controlpoint: 0%)
])
```

25% (default)



0%



Default: `25%`

edge-overshoot `ratio` or `float` or `length`

The overshoot on the left and right side of each wave.

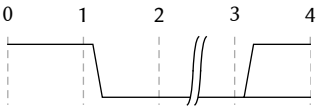
```
#let data = (signal: ((wave: "10|2"),))

*edge-overshoot: 0%* (default)
#dd.wave(data)

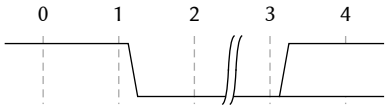
*edge-overshoot: 5mm*
#dd.wave(data,edge-overshoot: 5mm)

// 100% is equal to 7mm (symbol-width = 7mm)
```

edge-overshoot: 0% (default)



edge-overshoot: 5mm



Default: `0%`

wave-gutter `ratio` or `float` or `length`

The spacing between waves. Not included when an empty wave is inserted (wave = `(:)`).

Float and ratio values are treated as relative to symbol-width.

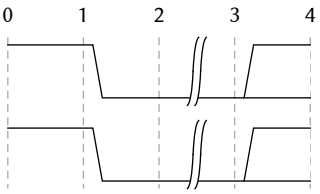
```
#let data = (signal: ((wave: "10|2"),(wave:
"10|2"),))

*wave-gutter: 4mm* (default)
#dd.wave(data)

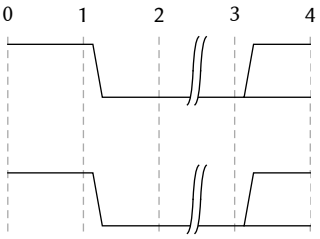
*wave-gutter: 100%*
#dd.wave(data, wave-gutter: 100%)

// 100% is equal to 7mm (symbol-width = 7mm)
```

wave-gutter: 4mm (default)



wave-gutter: 100%



Default: 4mm

name-gutter `ratio` or `float` or `length`

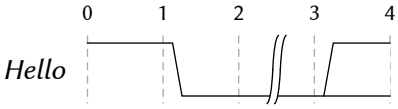
The spacing between wave and the wave name.

```
#let data = (signal: (
   (wave: "10|2", name: emph[Hello]),
 ))

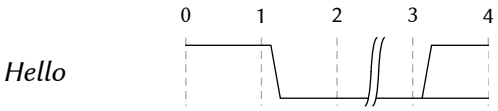
*name-gutter: 0.4* (default)
#dd.wave(data)

*name-gutter: 1.6*
#dd.wave(data, name-gutter: 1.6)
```

name-gutter: 0.4 (default)



name-gutter: 1.6



Default: 3mm

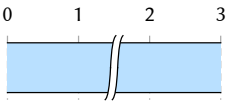
s-spacing `ratio` or `float` or `length`

Spacing between the $\int$ symbols of the time skip symbol.

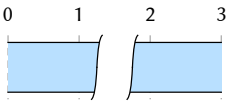
```
#let data = (signal: ((wave: "5|."),))

#stack(dir: ltr,[
  *s-spacing: 1mm* (default)
  #dd.wave(data)
],[
  *s-spacing: 50%*
  #dd.wave(data, s-spacing: 50%)
], spacing: 5mm)
```

s-spacing: 1mm (default)



s-spacing: 50%



Default: `1mm`

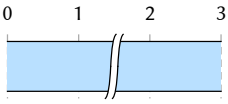
s-width `ratio` or `float` or `length`

Width of the $\int$ symbols of the time skip symbol.

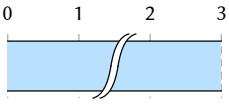
```
#let data = (signal: ((wave: "5|."),))

#stack(dir: ltr,[
  *s-width: 1.5mm* (default)
  #dd.wave(data)
],[
  *s-width: 50%*
  #dd.wave(data, s-width: 50%)
], spacing: 5mm)
```

s-width: 1.5mm (default)



s-width: 50%



Default: `1.5mm`

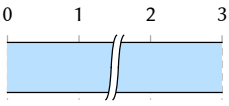
s-outside `ratio` or `float` or `length`

The «overshoot» of the $\int$ symbols of the time skip symbol outside of the wave height.

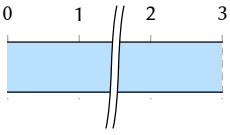
```
#let data = (signal: ((wave: "5|."),))

#stack(dir: ltr,[
  *s-outside: 1mm* (default)
  #dd.wave(data)
],[
  *s-outside: 50%*
  #dd.wave(data, s-outside: 50%)
], spacing: 5mm)
```

s-outside: 1mm (default)



s-outside: 50%



Default: `1mm`

mark-scale `float`

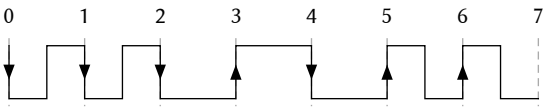
The size of the arrow marks applied to clocks (`"N"`, `"P"`) and logic signal flanks (`"L"`, `"H"`).

```
#let data = (signal: (
   (wave: "NNLHLPP"),
 ))

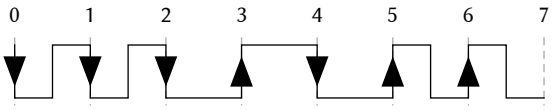
*mark-scale: 1* (default)
#dd.wave(data)

*mark-scale: 2*
#dd.wave(data, mark-scale: 2)
```

mark-scale: 1 (default)



mark-scale: 2



Default: `1`

stroke `stroke`

The styling of the solid wave strokes.

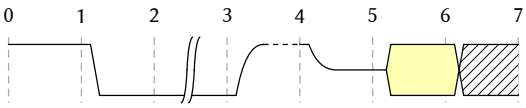
```
#let data = (signal: ((wave: "10|uz3x"),))

*Default*
#dd.wave(data)

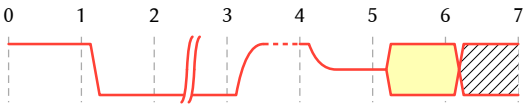
#raw("(paint: red, join: \"round\", cap:
\"round\")", lang: "typc")

#dd.wave(data, stroke: (paint: red, join:
"round", cap: "round"))
```

Default



`(paint: red, join: "round", cap: "round")`



Default: `(thickness: 0.5pt, cap: "square", paint: black)`

stroke-dashed `stroke`

The styling of the dashed strokes of the symbols `"u"` and `"d"`. Missing configurations are inherited from `stroke` (as seen in the example below).

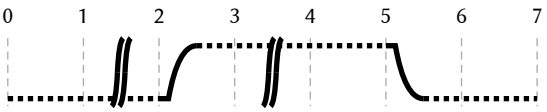
```
#let data = (signal: ((wave: "d|u|.d."),))

*Default*
#dd.wave(data, stroke: 2pt)

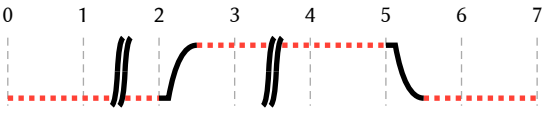
#raw("paint: red, dash: \"dotted\"", lang:
"typc")

#dd.wave(data, stroke: 2pt, stroke-dashed:
(paint: red, dash: "dotted"))
```

Default



`paint: red, dash: "dotted"`



Default: `(dash: (2pt, 1.75pt), cap: "butt")`

show-ticks `bool` or `function`

Controls the drawing of the tick lines and labels based on boolean conditions. `true` draws all tick lines and labels, `false` does not. Passing a function in the shape of `(n) => {..} -> bool` can be used to conditionally draw tick lines (e.g. only odd lines).

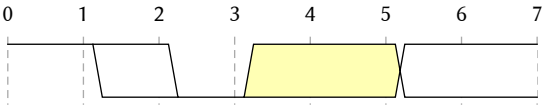
```
#let data = (signal: ((wave: "1203.2."),))

*show-ticks: `true`* (Default)
#dd.wave(data)

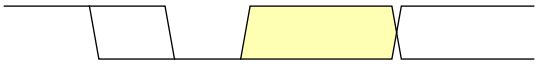
*show-ticks: `false`*
#dd.wave(data, show-ticks: false)

*show-ticks: `calc.odd`*
#dd.wave(data, show-ticks: (n) =>
calc.odd(n))
```

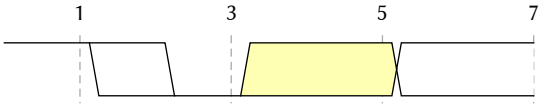
show-ticks: `true` (Default)



show-ticks: `false`



show-ticks: `calc.odd`



Default: `true`

tick-gutter `ratio` or `float` or `length`

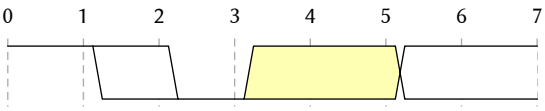
Spacing between the tick numbers and tick lines (including tick-overshoot). Length is relative to symbol-width.

```
#let data = (signal: ((wave: "1203.2."),))

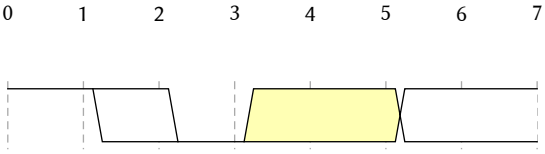
*tick-gutter: `20%`* (Default)
#dd.wave(data)

*tick-gutter: `8mm`*
#dd.wave(data, tick-gutter: 8mm)
```

tick-gutter: 20% (Default)



tick-gutter: 8mm



Default: 20%

tick-overshoot `ratio` or `float` or `length`

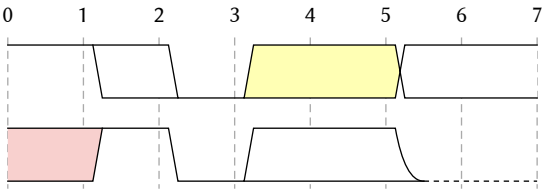
Overshoot of the tick line outside of the **total** diagram height. Length is relative to symbol-width.

```
#let data = (signal: ((wave: "1203.2."),
(wave: "9102.d.")))

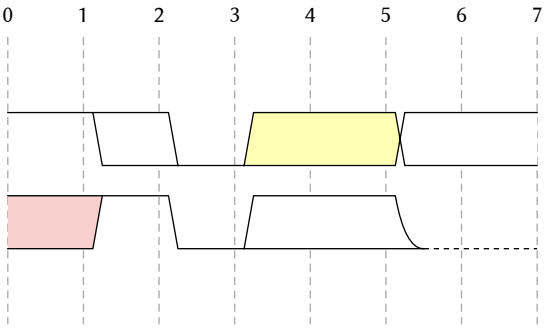
*tick-overshoot: `1mm`* (Default)
#dd.wave(data)

*tick-overshoot: `100%`*
#dd.wave(data, tick-overshoot: 100%)
```

tick-overshoot: 1mm (Default)



tick-overshoot: 100%



Default: 1mm

tick-format `none` or `function`

A rendering function with one parameter representing the tick number. Must return `content` and essentially describes how the label numbers are rendered. Setting it to `none` removes the labels.

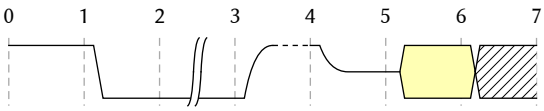
```
#let data = (signal: (
   (wave: "10|uz3x"),
 ))

*Default*
#dd.wave(data)

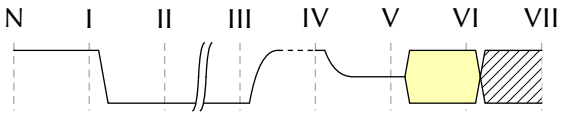
*tick-format:* `(n) => numbering("I", n)`
#dd.wave(data, tick-format: (n) =>
numbering("I", n))

*tick-format:* `none`
#dd.wave(data, tick-format: none)
```

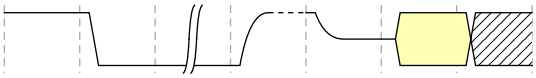
Default



tick-format: `(n) => numbering("I", n)`



tick-format: `none`



Default: `(n) => text(0.8em, numbering("1", n))`

tick-stroke `none` or `stroke` or `function`

The stroke of the tick lines. `none` disables the lines and passing a function in the shape of `(n) => stroke | none` allows for special styling based on tick number

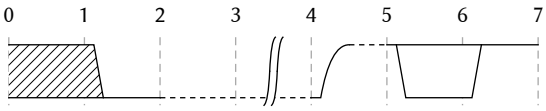
```
#let data = (signal: ((wave: "x0d|u=1"),))

*Default*
#dd.wave(data)

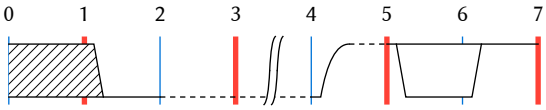
*tick-stroke: `function`*
#dd.wave(data, tick-stroke: (n) => {
  if calc.odd(n) {
    red + 2pt
  } else {
    blue + 0.5pt
  }
})

*tick-stroke:* `none`
#dd.wave(data, tick-stroke: none)
```

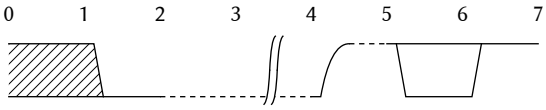
Default



tick-stroke: `function`



tick-stroke: `none`



Default: `(thickness: 0.5pt, paint: gray, dash: "densely-dashed")`

guide-stroke `none` or `stroke` or `dictionary`

When set to a stroke or a dictionary with stroke entries `top` and `bottom`, this will draw horizontal lines at the *Low* and *High* level of each wave.

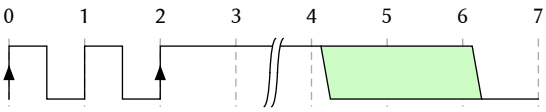
```
#let data = (signal: ((wave: "PpH|7.0"),))

*guide-stroke: `none`* (Default)
#dd.wave(data)

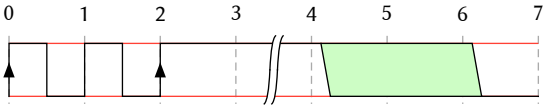
*guide-stroke: `stroke`*
#dd.wave(data, guide-stroke: red + 0.5pt)

*guide-stroke: `dictionary`*
#dd.wave(data, guide-stroke: (top: red +
0.5pt, bottom: blue + 0.5pt))
```

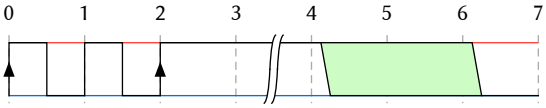
guide-stroke: none (Default)



guide-stroke: stroke



guide-stroke: dictionary



Default: none

name-format `function`

The formatting of the wave names, which are attached next to each respective wave.

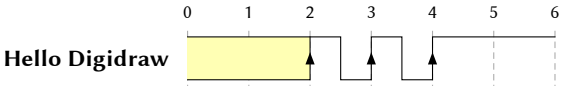
```
#let data = (signal: ((wave: "3.P.H.", name:
[Hello Digidraw]),))

*Default*
#dd.wave(data)

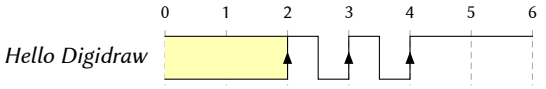
*name-format: `emph`*
#dd.wave(data, name-format: emph)

*name-format: `raw`* (name is `content` here)
#dd.wave(data, name-format: (name) =>
raw(name.text))
```

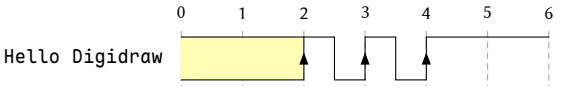
Default



name-format: emph



name-format: raw (name is content here)



Default: `name => text(1em, weight: "bold", bottom-edge: "baseline", name)`

data-format `function`

The formatting of the wave bus data/labels. Similar to name-format, with the exception that `data-format` also can use `#align` to align the labels in the respective bus.

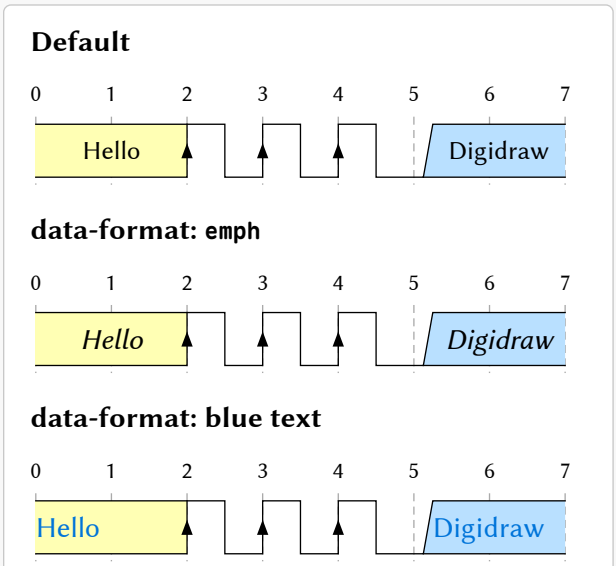
Use `debug: "labels"` to see the bounding boxes of where labels can be placed.

```
#let data = (signal: ((wave: "3.P..5.", data:
"Hello Digidraw"),))

*Default*
#dd.wave(data)

*data-format: `emph`*
#dd.wave(data, data-format: name =>
align(horizon+center, emph(name)))

*data-format: blue text*
#dd.wave(data, data-format: (name) =>
align(horizon,text(blue, name)))
```



Default:
```
data => align(center + horizon, text(
    0.9em,
    bottom-edge: "baseline",
    data,
  ))
```

bus-colors `dictionary`

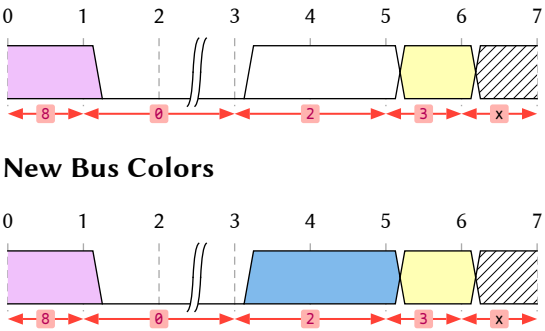
A dictionary containing the bus number/type as the key and fill colors, tilings or gradients as value.

Note When not all bus colors are configured, the remaining ones **will** be set to the default values.

```
#let data = (signal: (
  (wave: "80|2.3x"),)
)
#let bus-colors = ("2": blue.lighten(50%))

#dd.wave(data, debug: "symbols")

*New Bus Colors*
#dd.wave(data, debug: "symbols", bus-colors:
bus-colors)
```



`digidraw-x-pattern()` is the gray diagonal lines (as seen in the example below). It can be customized

Default: (

```
    "=": white,
    "2": white,
    "3": rgb("#ffffb4"),
    "4": rgb("#ffe0b9"),
    "5": rgb("#b9e0ff"),
    "6": rgb("#ccfdfe"),
    "7": rgb("#cdfdc5"),
    "8": rgb("#f0c1fb"),
    "9": rgb("#f8d0ce"),
    "x": digidraw-x-pattern(),
  )
```

others `cetz`

After the diagram is rendered, additional `cetz` elements can be added on top of it. The parameter accepts a function with following shape

```
(info) ⇒ {..}
```

where `info` consists of various information useful when placing elements (the values of these properties are given **in cetz units**):

```
(total-width, total-height, size-ref, symbol-height, wave-origins)
```

`total-width` is equal to the longest wave and can be read from the tick numbers
`total-height` is equal to the sum of all wave heights and gutters
`size-ref` (or `symbol-width`) is useful when you want to convert lengths to respective cetz units or in reverse.
`symbol-height` is equal to the symbols' height
`wave-origins` the actual points used to draw the waves (equal to the top left corner of a wave).

Use `debug: "coordinates"` to find specific points. Additionally you can use `set-origin(..)` to jump to specific points the diagram:

- `"diagram-origin"` is the top left corner of the diagram, indicated by the blue circle with a dot inside in the example.
- `"waveX"` where `X` is the 0-indexed number of the respective wave. Is equal to the **bottom left corner** of each wave, also indicated by the red dots in the example.

> **Caution**
> I'm currently not too happy about the implementation and it might change in the future.

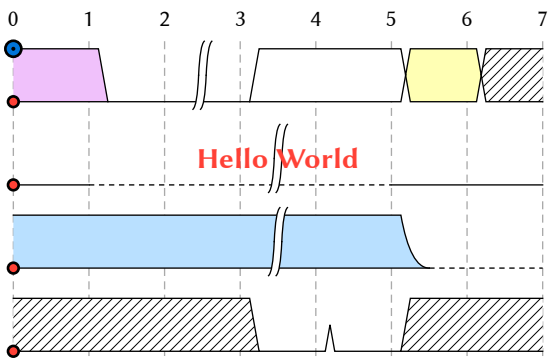
```
#import "@preview/cetz:0.5.2"

#dd.wave((signal: (
  (wave: "80|2.3x"), (wave: "0d.|.l."),
  (wave: "5..|.d."), (wave: "xxx00xx"),
)), others: (info) ⇒ {
  import cetz.draw: *

  set-origin("wave1")
  content((info.total-width/2,-info.symbol-
height/2),text(red, strong[Hello World]))

  for i in range(4) {
    set-origin("wave" + str(i))
    circle((0,0),radius: 2pt, fill: red)
  }

  set-origin("diagram-origin")
  circle((0,0),radius: 3pt, fill: blue)
  circle((0,0),radius: 0.5pt, fill: black,
stroke: 0.5pt)
})
```



Default: none

IV.2 digidraw-x-pattern

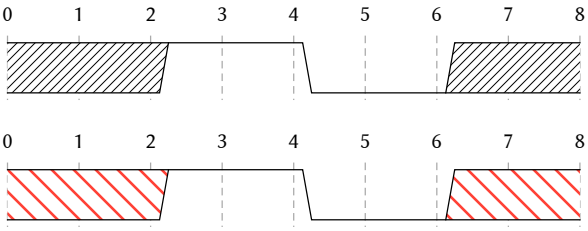
The default tiling/fill for the "x"-bus (but not limited to it). It can be customized with color (stroke), sizing (size) and direction (dir).

```
#let data = (signal: ((wave: "x.1.0.x."),))

*Default*
#dd.wave(data)

#dd.wave(data,
  bus-colors: (
    "x": dd.digidraw-x-pattern(
      stroke: red,
      size: 3mm,
      dir: ttb
    )
  )
)
```

Default



IV.2.a Parameters

```
digidraw-x-pattern(
  stroke: stroke,
  size: length,
  dir: direction
) → tiling
```

stroke `stroke`

Color of the diagonal lines

Default: `black + 0.3pt`

size `length`

Vertical or horizontal spacing of the lines. The tiling uses a square box, in which one line segment is drawn.

Default: `1.25mm`

dir `direction`

Direction of the diagonal lines.

- `btt` goes from bottom-left to top-right
- `ttb` goes from top-left to bottom-right

Default: `btt`

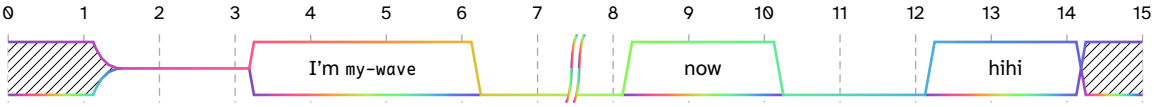
V \ Tips & Tricks

V.1 Pre-Configuring the #wave Function

To preconfigure a function or set a «Global» configuration, you can use the *with*-function to pre-apply parameters.

```
#let my-wave = dd.wave.with(stroke: gradient.linear(..color.map.rainbow))

#my-wave(
  (signal: (
    (wave: "xzz2..0|2.0.2.x",data: ([I'm `my-wave`],[now],[hihi],)),)
  )
)
```



V.2 Making Diagrams Referenceable

To make timing diagrams referencable, enclose it in a `#figure` element. Additionally, to make labelling (`...<a-label>`) work, define a new function or «overload» the `#wave` function:

```
#let wave(..args, caption: none) = figure(
  dd.wave(
    ..args
  ),
  caption: caption,
  supplement: [Diagram],
  kind: "timing-diagrams"
)

#wave(
  (signal: (
    (wave: "x12.udx34z.5.x.",data: "Hello This is Digidraw"),)
  ),
  caption: [Use case of a timing diagram.]
) <dd:example>

When a need to reference the @dd:example occurs, `@dd:example` can be used. And for listing the
diagrams, the outline target `figure.where(kind: "timing-diagrams")` is to be used:

```typst
#outline(target: figure.where(kind: "timing-diagrams"))
```
```

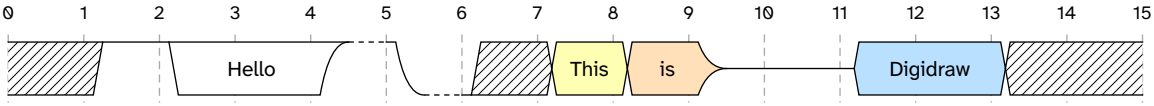


Diagram 1: Use case of a timing diagram.

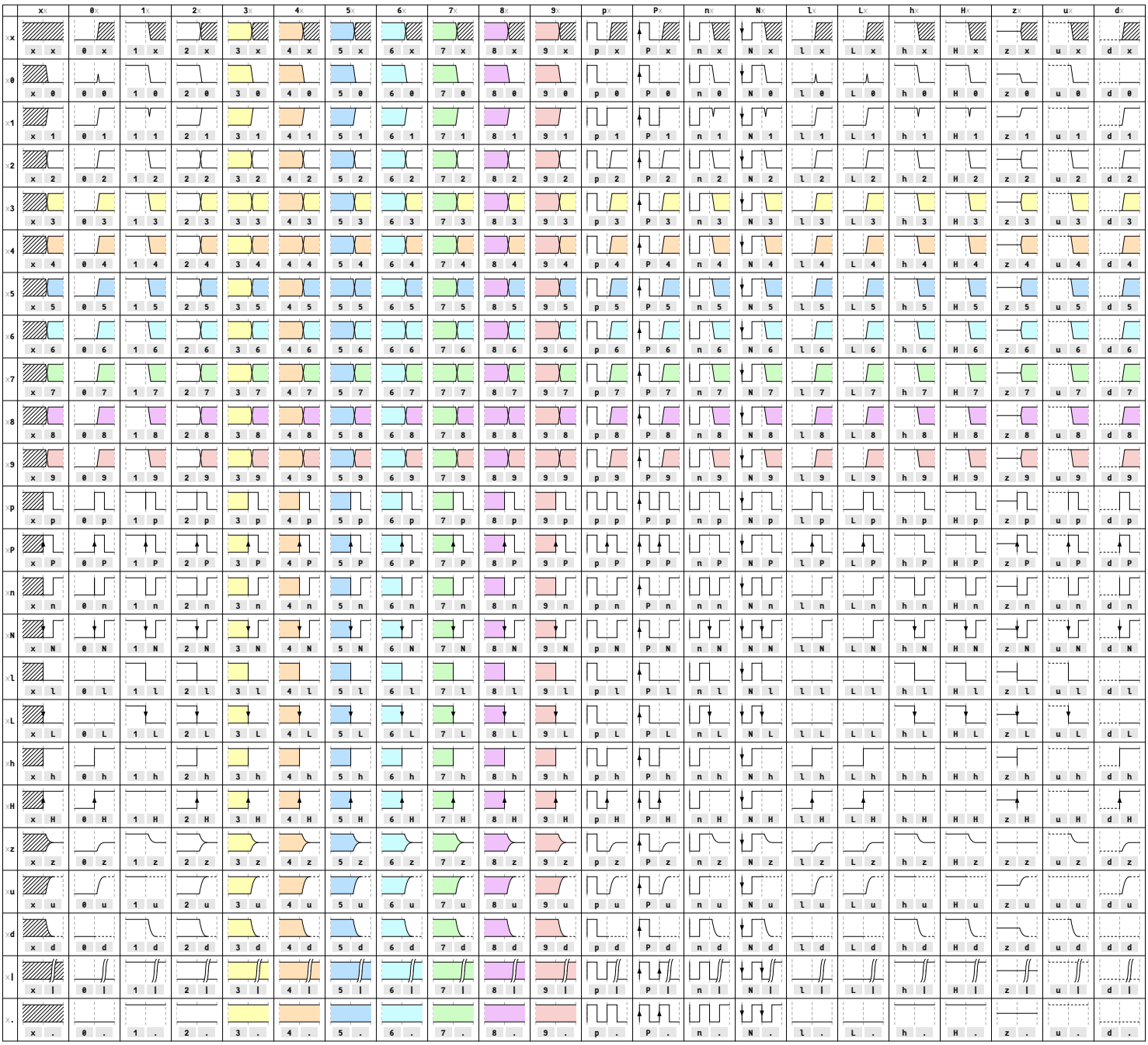
When a need to reference the Diagram 1 occurs, `@dd:example` can be used. And for listing the diagrams, the outline target `figure.where(kind: "timing-diagrams")` is to be used:

```
#outline(target: figure.where(kind: "timing-diagrams"))
```

VI \ Symbol Matrix

The symbol matrix shows all symbol combinations. This is also how the system is debugged (to some degree). If you find any wrong combinations, please report them at https://codeberg.org/joelvonrotz/typst-digidraw!

Due to the sheer size of the table, it's rather small. So it's recommended to use a PDF viewer instead of a printed version. Or checkout the matrix as a standalone document over at https://codeberg.org/joelvonrotz/typst-digidraw/src/tag/0.9.3/docs/symbol_matrix.pdf,



VII \ Changelog

Version 9.0 – *Initial release of `digidraw`*

- Implemented the function `#wave` which supports the main parts of the WaveDrom syntax.
- Added various customization settings to the `#wave` function to allow cool styling

Version 9.1

- Typo fixes

Version 9.2

- **Merge-Request by ALVAROPING1**
  Expanded `stroke-tick-lines` to support a function with one parameter indicating the tick number. As a return value, either a dictionary (containing the stroke parameters) or a stroke is expected. This allows for styling tick lines at odd or even tick numbers
- Inspired by ALVAROPING1's feature, the tick line system has been expanded to also allow function based visibilty and stroke style reseting for specific lines
- fixed some symbol transitions (transition from `z` symbol to `p`,`P`,`n`,`N`,`l`,`L`,`h`,`H`)
- added one new example to show of the new features
- updated manual to be on par with package version 0.9.2

Version 9.3

Highlight

- Internal refactoring of the `#wave` function
  - almost all symbols have now an upper part, a lower part and a body. Upper parts are merged together, leading to one long line, which then are decorated with the lower parts. The body is added behind these parts (see buses).

Added

- Added `edge-overshoot` parameter to `#wave` to allow symbol lines to go over the wave edges (left and right)
- Added `tick-gutter` parameter, that defines the space between the tick numbers and the tick line (including `tick-overshoot`)
- Added `step3` parameter, that controls the end point of the bezier curves (i.e. symbols `"z"` or `"d"`)
- Added `bezier-controlpoint` parameter, that controls the beziers' control point (to make the curve ; might change the name at some point)
- Added `s-spacing`, `s-width` and `s-outside` parameters, that control the shape of the s-symbols of a time delay (symbol `"|"`)
- Added `others` parameter, which allows for additional cetz content to be added on top of the diagram. Using `"coordinates"` debug mode helps with finding the correct coordinates.

Changed

- `symbol-width` is now the size reference of the diagram
- Renamed `wave-height` to `symbol-height` and changed allowed value types
- Renamed `stroke-guides` to `guide-stroke` and added support for styling upper and lower guide lines
- Renamed `show-tick-lines` to `show-ticks`
- Renamed `stroke-tick-lines` to `tick-stroke`
- Renamed `inset-1` to `step1`
- Renamed `inset-2` to `step2`

- Split Debugging into four parts: `"labels"`, `"symbols"`, `"steps"` and `"coordinates"` (multiple can be activated)
- Updated docs to reflect the changes

⊖ Removed

- Removed `wave-width` parameter, because I currently don't have an idea how to implement this with the new system. Plus, the user should be smart enough on how to limit the widths of a diagram by changing the wave strings.
- Removed `show-guides` parameter
- Removed `debug-offset` parameter (I might reimplement this at some point)
- Removed `size` parameter, as `symbol-width` represents this parameter
- Removed ability to reset configurations by passing `auto` to the respective parameter